ALPHCOM CUSTOMER CHURN PREDICTION

Capstone Project Report

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1. Understanding the Business Problem

1.1 Problem Statement

AlphaCom operates in a highly competitive telecommunications environment where customers face minimal barriers to switching service providers.

Over recent periods, the company has observed a noticeable increase in customer churn, which has begun to erode recurring revenues, customer lifetime value, and longterm profitability.

Churn decisions are rarely driven by a single factor; instead, they emerge from a combination of contract flexibility, perceived pricing fairness, service experience, and the length of time a customer has remained with the company.

In the absence of predictive insight, AlphaCom's retention efforts tend to be reactive, limiting the organization's ability to intervene at an early stage.

1.2 Need for the Project

Industry benchmarks suggest that churn rates in the telecommunications sector typically fall in the range of 20–30%, making customer retention a strategic priority rather than a purely operational concern.

Retaining an existing customer is substantially less expensive than acquiring a new one, particularly in mature and highly saturated markets. Without a structured, data-driven churn prediction framework, AlphaCom risks inefficient use of retention budgets, overreliance on blanket discounts, and missed opportunities to engage customers before disengagement becomes irreversible.

1.3 Business Opportunity

By systematically analyzing historical customer data and applying predictive modeling techniques, AlphaCom can move from reactive churn management toward a proactive, insight-led retention strategy. An effective churn prediction model allows the business to prioritize high-risk customers, design targeted and value-aligned retention offers, improve customer experience at critical lifecycle stages, and ultimately strengthen customer lifetime value while maintaining control over retention costs.

1.4 Defining Success

Success for this interim phase is defined as developing an interpretable baseline churn prediction model that can reliably predict customer churn risk and identify the key business drivers of churn. The focus at this stage is on insight generation and decision support rather than maximum predictive performance.

2. Data Overview

The dataset used in this study contains 12,055 individual customer records and 20 variables spanning customer demographics, service subscriptions, billing characteristics, and contract details. The target variable, Churn, indicates whether a customer has discontinued services. Taken together, these variables provide a comprehensive view of the customer lifecycle, from service adoption through billing behavior and contract commitment, making the dataset well suited for churn analysis.

Table 1 Dataset variable description

Category	Key Variables	Description
Target	Churn	Binary indicator of customer attrition
Demographics	Gender, SeniorCitizen, Partner, Dependents	Customer profile attributes
Services	PhoneService,InternetService, OnlineSecurity, TechSupport	Subscribed products and valueadded services
Account	Contract, PaperlessBilling, PaymentMethod	Contractual and billing preferences
Usage/Cost	Tenure, MonthlyCharges, TotalCharges	Lifecycle duration and financial value

Table 2 Data overview

Column	***Non-Null Count***	***Dtype***	***Notes / Meaning***
gender	*12055*	*object*	*Customer's gender (Male/Female)*
SeniorCitizen	*12055*	*int64*	*Indicator (0 = No, 1 = Yes)*
Partner	*12055*	*object*	*Whether the customer has a partner (Yes/No)*
Dependents	*12055*	*object*	*Whether the customer has dependents (Yes/No)*
tenure	*11451*	*float64*	*Number of months the customer has stayed*
PhoneService	*12055*	*object*	*Whether the customer has phone service (Yes/No)*
MultipleLines	*12055*	*object*	*Whether the customer has multiple phone lines*
InternetService	*12055*	*object*	*Type of internet service (DSL/Fiber optic/No)*
OnlineSecurity	*12055*	*object*	*Whether the customer has online security add-on*
OnlineBackup	*12055*	*object*	*Whether the customer has online backup add-on*
DeviceProtection	*12055*	*object*	*Whether the customer has device protection added*
TechSupport	*12055*	*object*	*Whether the customer has tech support add-on*
StreamingTV	*12055*	*object*	*Whether the customer has streaming TV service*
StreamingMovies	*12055*	*object*	*Whether the customer has streaming service*
Contract	*12055*	*object*	*Contract type (Month-to-month/One year/Two year)*
PaperlessBilling	*12055*	*object*	*Whether the customer uses paperless billing*
PaymentMethod	*12055*	*object*	*Payment method*
MonthlyCharges	*12055*	*object*	*Monthly charges billed to the customer*

TotalCharges	*12055*	*object*	*Total charges billed to the customer*
Churn	*12055*	*object*	*Target variable: whether the customer churned*

---

3. Data Quality Issues and Preprocessing

3.1 Methodology

The data preprocessing stage focused on improving data quality while preserving realistic business behavior.

The following steps were taken to prepare the data:

- Duplicates: 24 records removed.
- Anomalies: Negative tenure values dropped.
- Feature Selection: TotalCharges will be dropped during model buliding
- Splitting: 70/30 Stratified Split.

3.2 Cleaning Numeric Columns

- Non-numeric characters were removed from *MonthlyCharges* and *TotalCharges*
- Columns were converted to numeric format

3.3 Data Type Conversion

- Boolean variables converted to logical flags
- Categorical variables standardized (lowercase, trimmed) to remove duplicate categories
- Converted to categorical data types for efficient modeling

3.4 Missing Value Treatment

Missing and invalid values were handled using business-driven logic rather than purely statistical imputation. Negative tenure values were treated as invalid and set to missing, as they are not logically meaningful in a customer lifecycle context. Absolute values were not taken since the root cause of the error is unknown.

MonthlyCharges was imputed using median values derived from customers with similar service profiles, preserving realistic pricing behavior. TotalCharges, a cumulative billing variable influenced by prorations and one-time adjustments, was retained for exploratory analysis but excluded from modeling to prevent noise, multicollinearity, and data leakage.

This approach balances data completeness with analytical integrity.

3.5 Class Imbalance Handling

The observed churn distribution reflects real-world business conditions rather than sampling bias. As a result, no resampling techniques were applied at the baseline stage.

Instead, evaluation metric AUC-PR was selected to explicitly prioritize the correct identification of churn-prone customers, aligning model evaluation with business objectives.

3.6 Outlier Assessment and Treatment

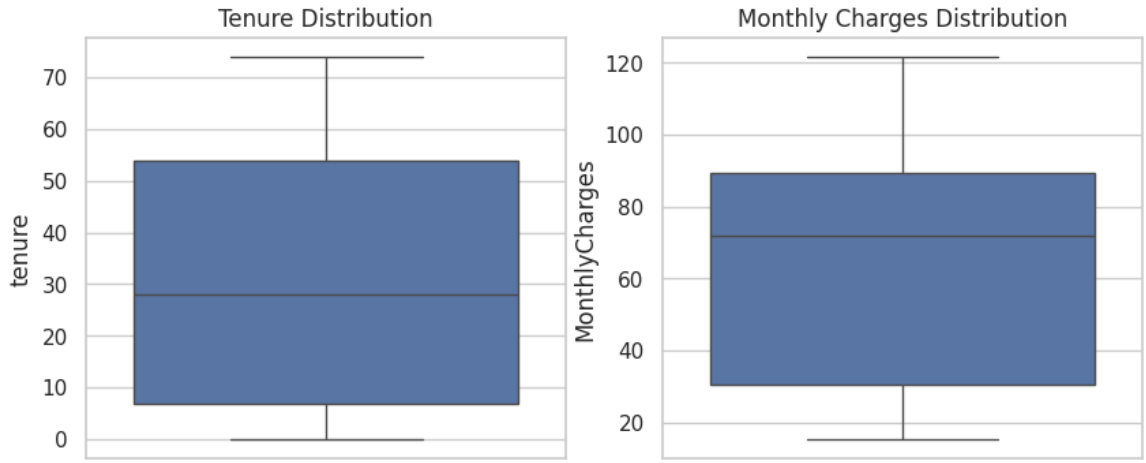

Figure i Outlier check

1. IQR based detection method was used to detect outliers
2. No statistically significant outliers were found
3. No rows were dropped

The absence of statistically significant outliers in tenure and MonthlyCharges indicates strong underlying data quality. This validation supports the robustness of downstream modeling and eliminates the need for aggressive outlier removal, thereby preserving valuable customer information while maintaining model stability.

3.7 Feature Scaling

Continuous variables ('tenure','MonthlyCharges') were standardized after the train–test split to ensure balanced contribution to the linear baseline model and to prevent scaledriven bias.

3.8 Data Leakage Prevention

1. All preprocessing (imputation, scaling, encoding) was fit on training data and applied to test data only.
2. Target variable never used in feature engineering.
3. To avoid data leakage, cumulative billing variables such as TotalCharges were excluded from the modeling dataset.
4. All preprocessing decisions relied solely on information available at the time of prediction, ensuring unbiased model evaluation.

3.9 Impact of Data Cleaning and Preprocessing

Prior to preprocessing, the dataset exhibited multiple quality issues that would have materially degraded model reliability and business conclusions, including invalid tenure values, non-numeric billing fields, inconsistent categorical labels, duplicate records **and** missing financial information. These deficiencies introduced noise, inflated model variance, and increased the risk of biased estimation and data leakage. Through a structured preprocessing pipeline, these issues were systematically resolved: invalid records were removed, billing fields were converted into consistent numeric formats, categorical attributes were standardized and encoded, duplicates were eliminated **and** missing MonthlyCharges values were imputed using median values from comparable customer profiles**.** Outliers in continuous variables were checked using IQR-based formula and it was found the data set is well behaved with regards to outliers hence no rows were dropped for outlier treatment. High-risk cumulative variables such as TotalCharges were intentionally excluded from modeling to prevent multicollinearity and leakage. The resulting dataset was clean, stable, and information-rich, substantially improving model convergence, generalization, and predictive performance, as reflected in the consistent gains in AUC-PR, recall, and F1-score across all advanced models.

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4. Exploratory Data Analysis (EDA)

4.1 EDA Methodology

Exploratory Data Analysis was undertaken to gain an initial understanding of data structure, quality, and customer behavior patterns. Univariate techniques were used to examine the distribution of individual variables, bivariate analyses explored relationships between churn and key explanatory features, and multivariate analyses were applied to understand interaction effects across multiple drivers. Throughout this process, emphasis was placed on business interpretability and practical relevance.

4.2 Data Quality Observations

Initial inspection revealed several data quality issues, including numeric variables stored as text, missing tenure values, and logically invalid entries likely arising from billing system or data-entry limitations. Addressing these issues prior to EDA and Modeling was essential to ensure that subsequent insights and predictions remained reliable.

4.3 Univariate Analysis

Univariate analysis shows that approximately 27.9% of customers in the dataset have churned, indicating a moderate but realistic level of class imbalance. Customer tenure exhibits a right-skewed distribution, with a large proportion of customers concentrated in the early months of service. Monthly charges vary widely across customers, reflecting differences in service configurations and pricing plans.

Interpretation: The high density of customers at low tenure levels suggests that AlphaCom faces elevated churn risk at the moment requiring immediate action. At the same time, wide variation in monthly charges indicates heterogeneous pricing sensitivity, implying that a one-size-fits-all retention strategy is unlikely to be effective.

Figure 2 and Figure 3 visually summarize these distributional patterns.

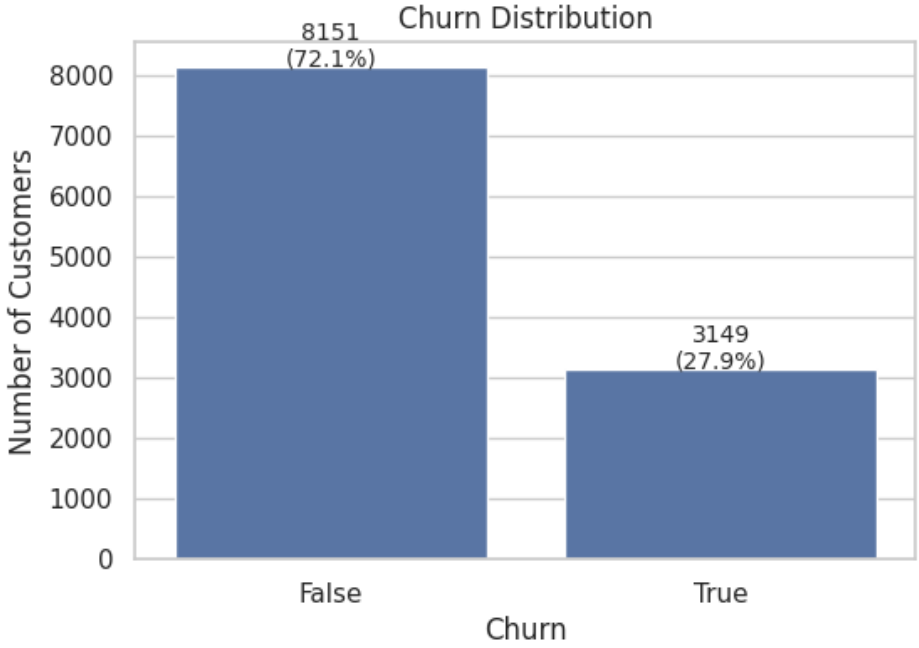

Figure 1 Distribution of Churn

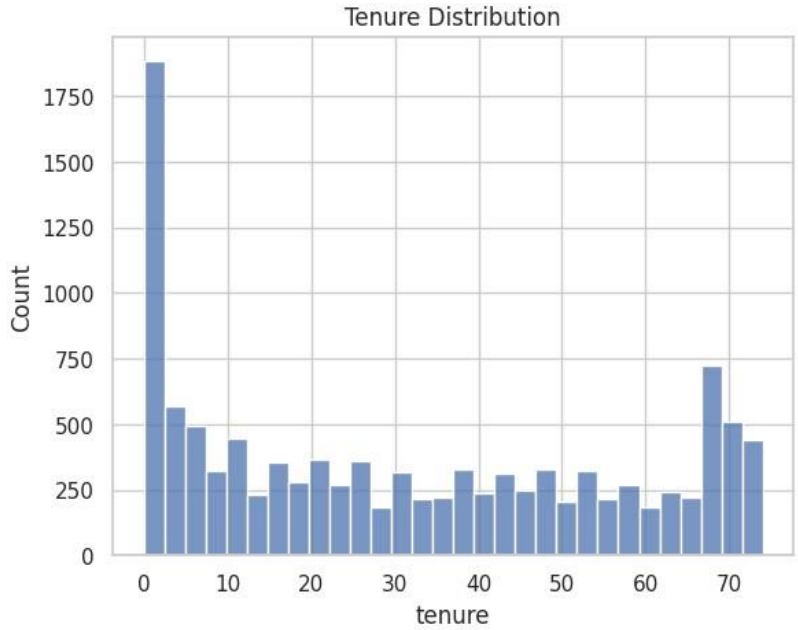

Figure 2 Distribution of Customer Tenure (Months)

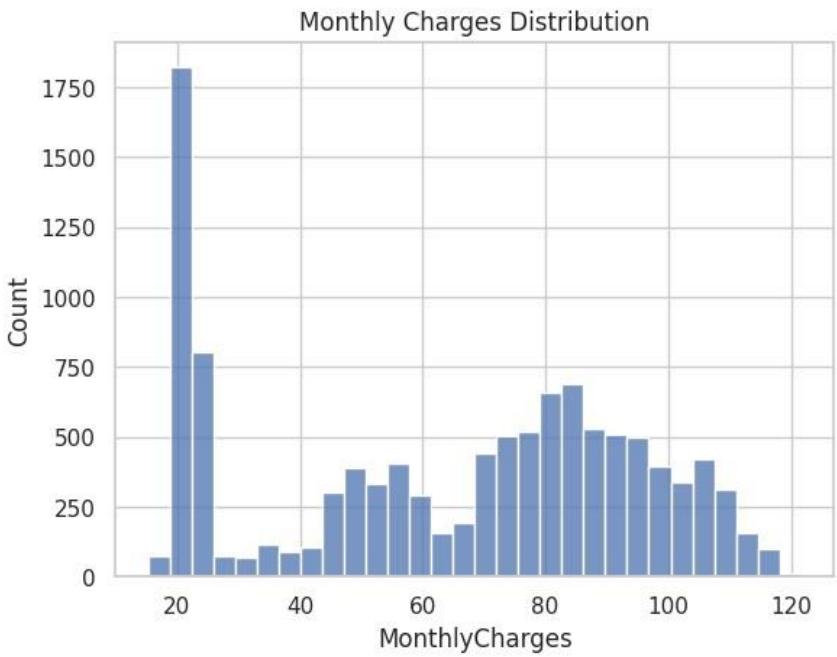

Figure 3 Distribution of Monthly Charges

4.4 Bivariate Analysis

Bivariate analysis highlights a strong inverse relationship between tenure and churn, with customers in the early stages of their relationship exhibiting significantly higher churn rates. Contract type emerges as a key driver, as customers on month-to-month contracts churn at much higher rates than those on longer-term agreements. In addition, higher monthly charges are associated with an increased likelihood of churn, suggesting pricing sensitivity. Customers also respond positively to value added services like Tech Support and Online Security.

Business Interpretation: These findings indicate that churn is most acute among customers who:

- Have not yet established long-term commitment
- Face higher recurring costs
- Retain contractual flexibility
- Facing issues with certain payment systems, namely electronic check
- Customer value add-on services like Tech Support and Online Security

This highlights the importance of early engagement strategies, pricing and billing transparency, and contract-based retention mechanisms.

Figures 4–9 illustrate these relationships.

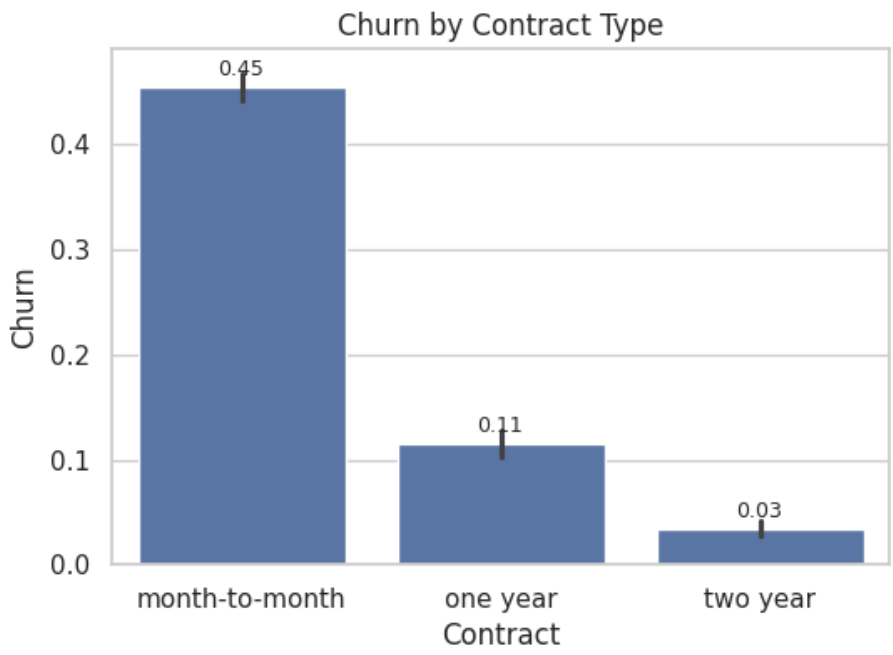

Figure 4 Churn Rate by Contract Type

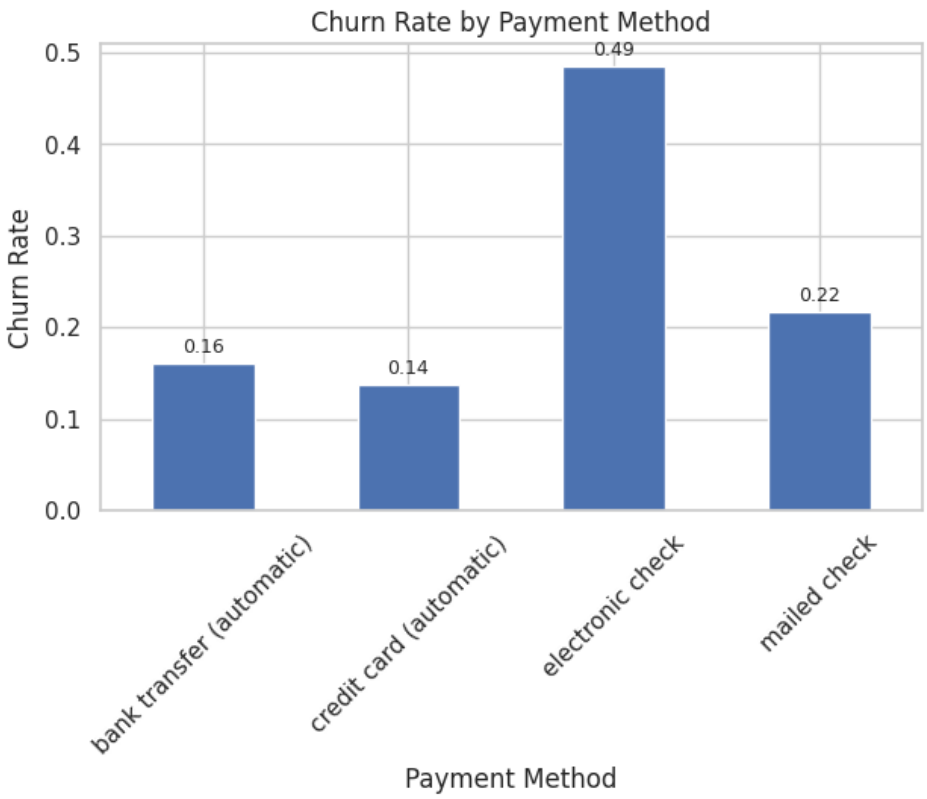

Figure 5 Churn Rate by Payment Method

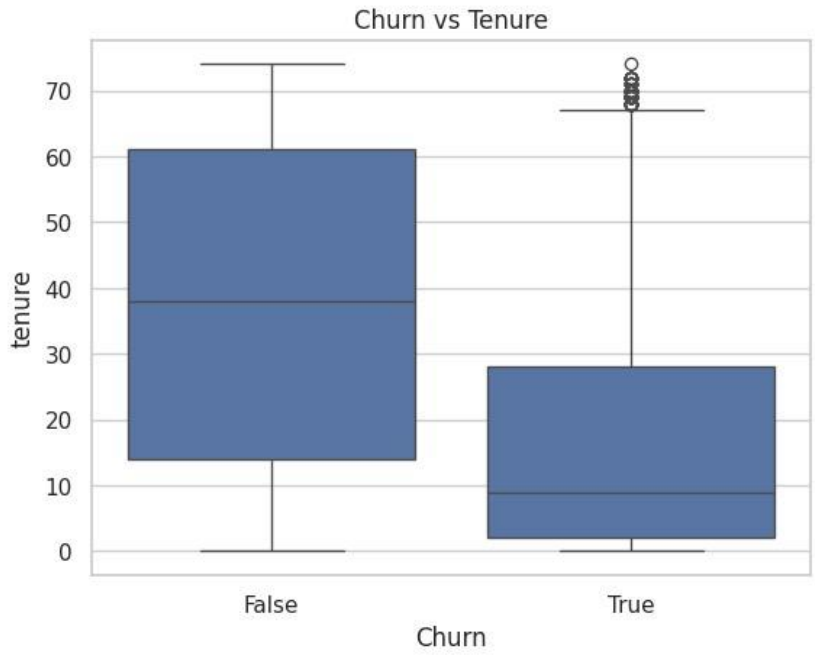

Figure 6 Tenure Distribution by Churn Status

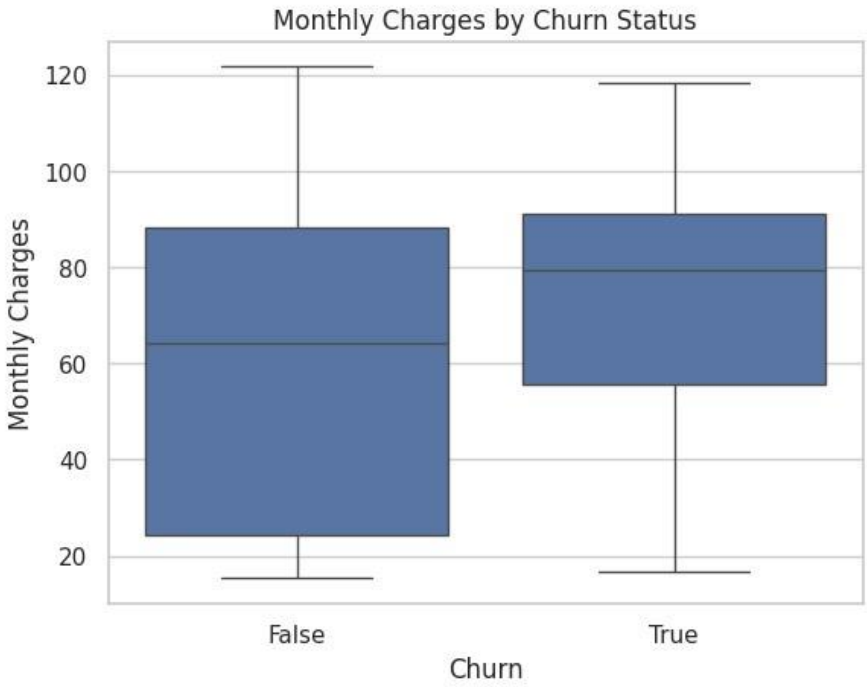

Figure 7 Monthly Charges Distribution by Churn Status

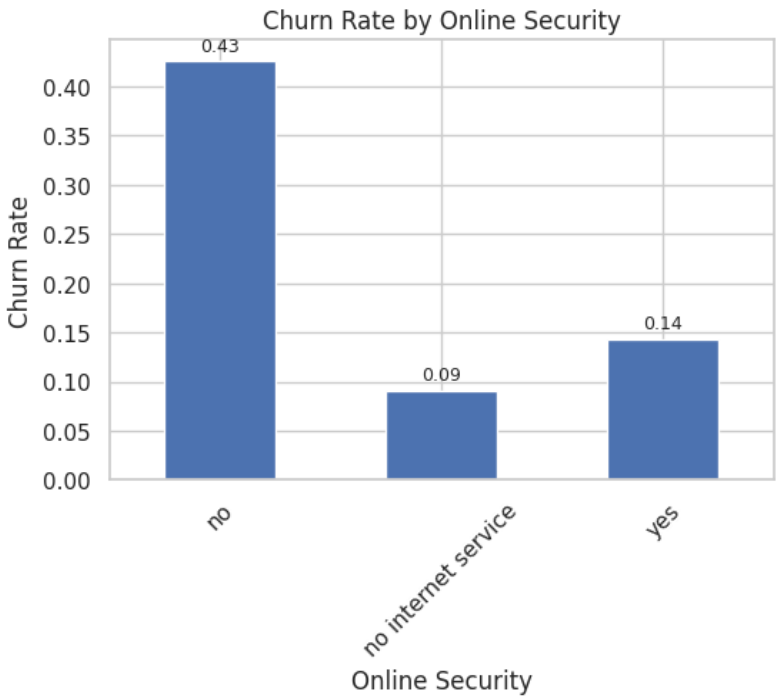

Figure 8 Churn rate by OnlineSecurity

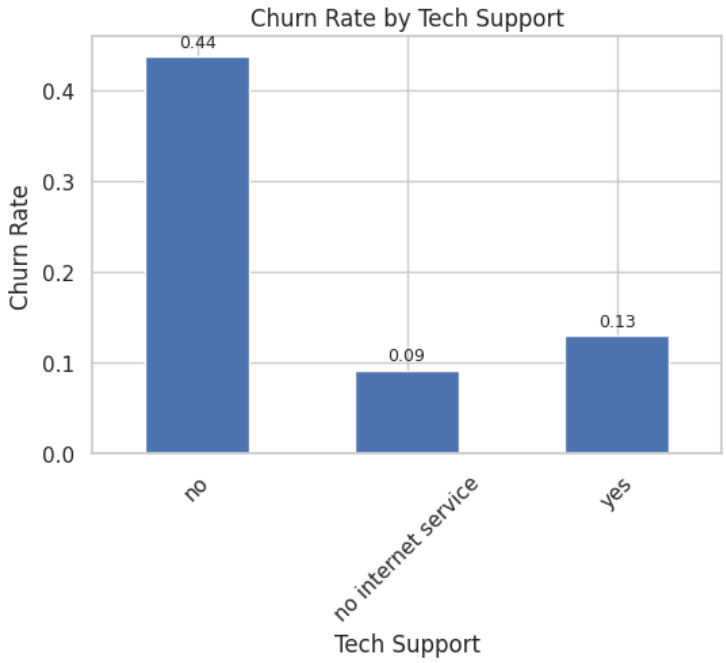

Figure 9 Churn Rate by Tech Support Subscription

4.5 Multivariate Analysis

Multivariate analysis reveals that churn risk is driven by the interaction of multiple factors rather than isolated variables. Customers with short tenure, month-to-month contracts, and higher monthly charges represent the highest-risk segment.

Conversely, customers subscribed to value-added services such as TechSupport and OnlineSecurity exhibit lower churn rates, suggesting that perceived service value and support quality play a protective role.

Strategic Insight: Churn is fundamentally a value-perception problem. Customers are more likely to remain when contractual commitment, pricing, and service quality jointly reinforce perceived value.

See Figures 10-12 below

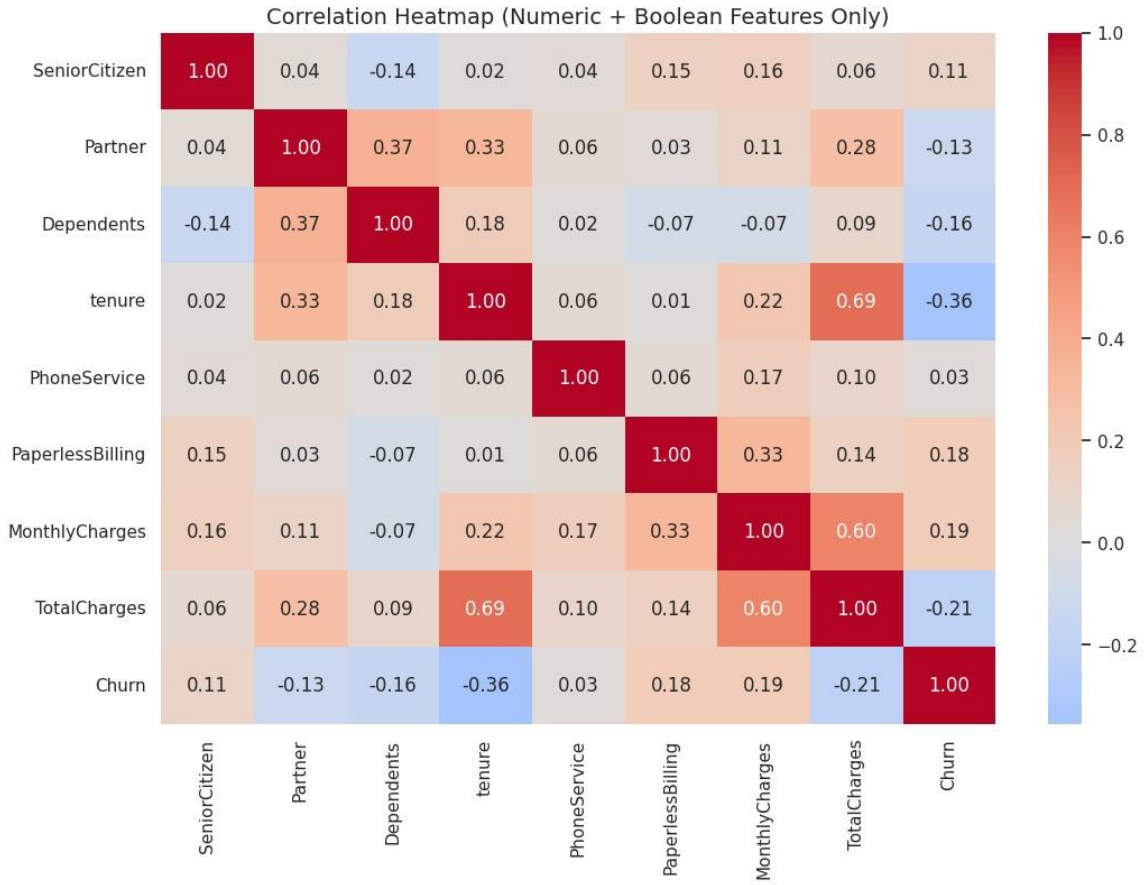

Figure 10 Feature Correlation Heatmap

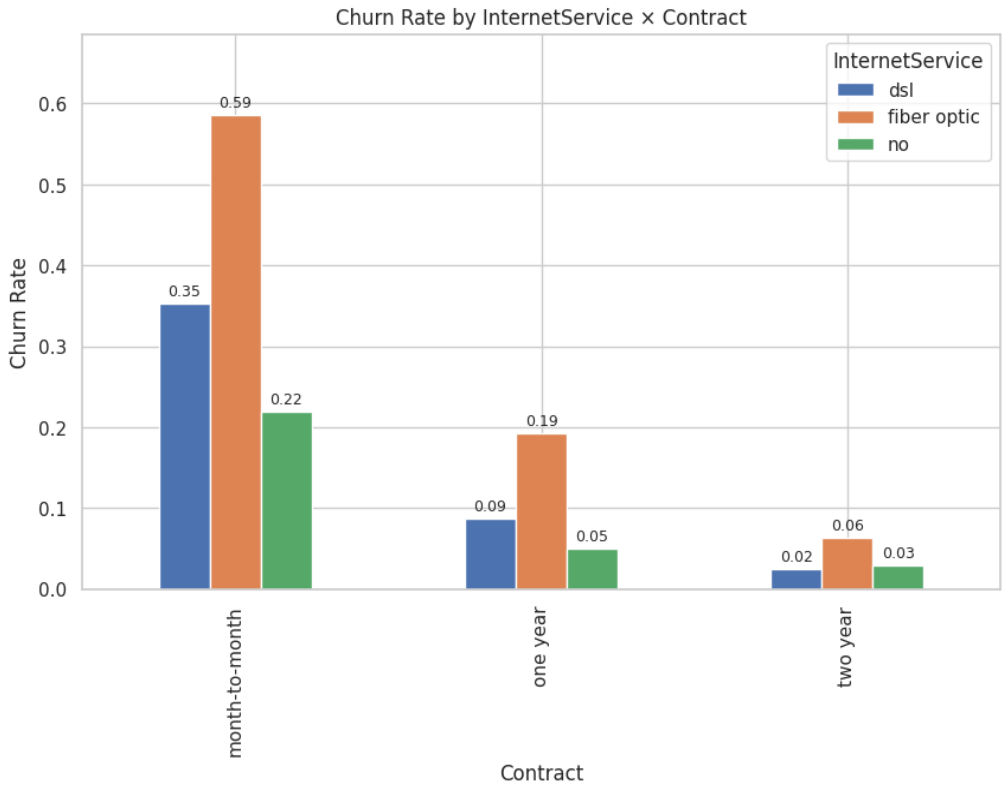

Figure 11 Interaction of Contract and Internet Service on Churn

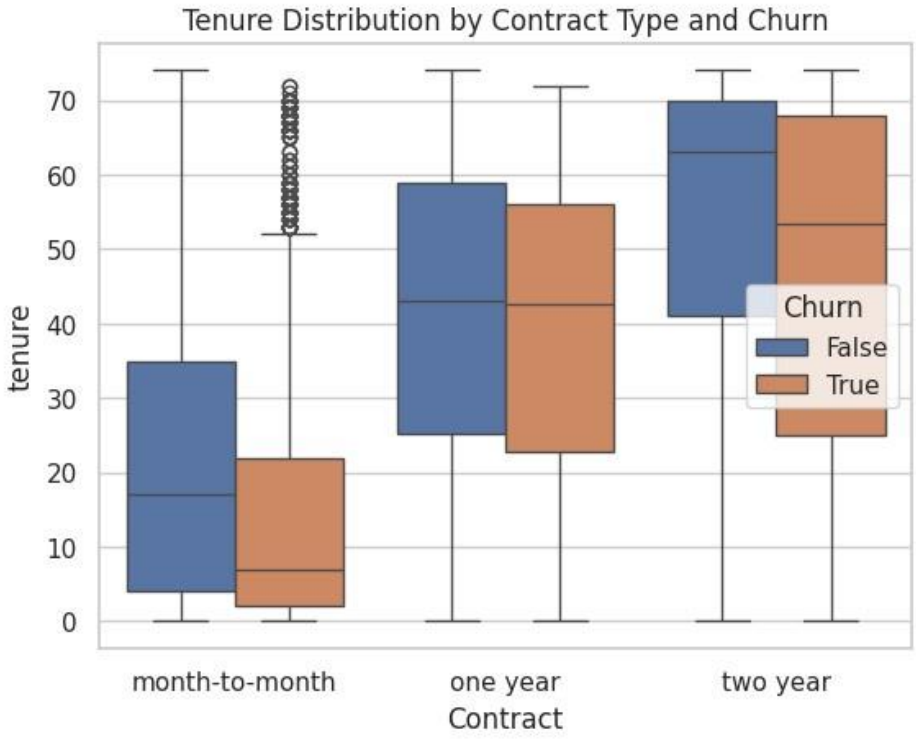

Figure 12 Tenure Distribution by Contract and Churn Status

4.6 Key EDA Insights Summary

- Churn is disproportionately concentrated within the first year of tenure
- Contract type is the most influential churn driver
- Higher monthly charges increase churn probability
- Value-added services significantly reduce churn risk
- Churn behavior reflects the interaction of pricing, commitment, and service experience

These insights directly inform feature selection and modeling decisions in subsequent stages.

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5. Model Building - Baseline Model

5.1 Model Selection Rationale

Logistic Regression was selected as the baseline model due to its interpretability, robustness, and suitability for binary classification problems. As a linear model, it provides transparent coefficient estimates that allow stakeholders to understand the directional impact of key drivers on churn probability.

Establishing a strong, interpretable baseline also provides a benchmark against which more complex models can be evaluated in later project phases.

5.2 Metric Selection

Missing a churn-prone customer is significantly more costly than incorrectly targeting a retained customer so the following were chosen:

Primary Metric: AUC-PR AUC-PR
was chosen because:

- Churn prediction involves class imbalance
- It focuses on identifying true churners
- It provides a realistic performance assessment for ranking high-risk customers

Secondary Metrics

- ROC-AUC: For overall discrimination assessment
- Recall: To minimize missed churners

5.3 Performance Metrics

Table 3 Baseline Model Performance Metrics

Metric	Value	Business Interpretation
Accuracy	0.7661	Overall correctness of predictions.
Precision	0.5549	Of those predicted to churn, 55% actually left.
Recall	0.8127	Crucial: Captures 81% of all churners.
F1-Score	0.6595	Balanced harmonic mean of Precision and Recall.
AUC-PR	0.6770	Model capability focused on the minority class (Churn).
ROC-AUC	0.8527	Model capability focused on overall class seperation

The Precision-Recall curve below illustrates the trade-off between Precision and Recall. The Area Under the Curve (AUC-PR) is the primary metric of choice for this imbalanced dataset.

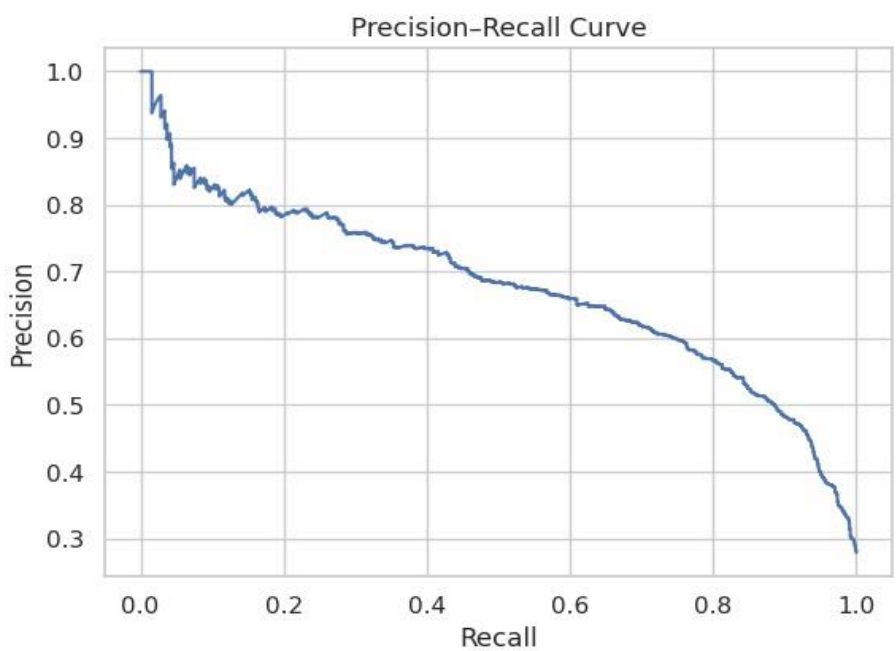

Figure 13 Precision-Recall Curve

5.4 Business Interpretation of Model Results

The baseline model demonstrates a strong ability to rank customers by churn risk, particularly in identifying high-risk segments early in the customer lifecycle. While the model is not optimized for precision at this stage, its recall, ROC-AUC and AUC-PR

performance align well with AlphaCom's retention objectives, where the cost of missing a churn-prone customer exceeds the cost of engaging a retained one.

From a business perspective, the baseline model gives us a reference point to compare advanced models performance on :

- Prioritizing retention outreach
- Supporting targeted marketing campaigns
- Informing contract and pricing strategy decisions

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6. Model Building – Advanced Models

6.1 Rationale for advanced models

While the baseline Logistic Regression model provides interpretability and strong initial performance, churn behavior is inherently nonlinear and driven by complex interactions between customer tenure, pricing, contract structure, and service usage. To capture these relationships more effectively, a set of advanced machine learning models was utilised and evaluated.

The following models were implemented:

- Decision Tree
- Random Forest
- AdaBoost
- Gradient Boosting
- XGBoost

These models were selected because they progressively increase model capacity and are well-suited for structured tabular data with mixed feature types and nonlinear relationships.

6.2 Performance of advanced models

The performance of all advanced models was evaluated using a comprehensive set of metrics to ensure both predictive strength and business relevance. Given the moderate class imbalance in the churn dataset, **AUC-PR was selected as the primary metric**, with **Recall and ROC-AUC** serving as secondary metrics to capture churner identification quality and overall discrimination respectively.

Table I Advanced Models performance metrics

Model	Best Threshold	Accuracy	Precision	Recall	F1 Score	AUC-PR	ROC-AUC	Training Time (sec)
Gradient Boosting	0.303	0.7841	0.5833	0.7894	0.6709	**0.6857**	0.8549	3.396
XGBoost	0.304	0.7764	0.5722	0.7841	0.6616	**0.6837**	0.8533	0.387
Random Forest	0.454	0.7755	0.5677	0.8159	0.6696	**0.6776**	0.8528	3.828
Logistic Regression	0.582	0.7894	0.5962	0.7577	0.6673	**0.6770**	0.8527	0.173
AdaBoost	0.402	0.7864	0.5932	0.7439	0.6601	**0.6478**	0.8441	1.687
Decision Tree	0.553	0.7705	0.5643	0.7757	0.6533	**0.6309**	0.8396	0.197

Primary Metric — AUC-PR

AUC-PR directly reflects the model's ability to correctly rank churn-risk customers, which is the core business objective of this project.

- Gradient Boosting achieved the highest AUC-PR (0.6857) among all advanced models.
- XGBoost followed closely with 0.6837, confirming strong ranking capability.
- Random Forest produced a solid but lower AUC-PR of 0.6776.
- AdaBoost (0.6478) and Decision Tree (0.6309) lagged behind, indicating weaker performance in prioritizing churn-risk customers.

This demonstrates that ensemble boosting methods significantly outperform simpler models when the objective is correct churn ranking.

Secondary Metric — Recall (Churn Capture)

Recall is critical from a business perspective because failing to identify churners carries high financial cost.

- Random Forest achieved the highest recall (0.8159), capturing the most churners.
- Gradient Boosting (0.7894) and XGBoost (0.7841) followed closely, maintaining strong churn detection.
- Logistic Regression (0.7577) and AdaBoost (0.7439) showed moderate performance.
- Decision Tree (0.7757) captured churners reasonably well but underperformed on overall ranking and stability.

This confirms that Gradient Boosting and XGBoost provide an excellent balance between churn capture and ranking precision.

Secondary Metric — ROC-AUC (Overall Discrimination)

ROC-AUC measures the model's ability to separate churners from non-churners across all thresholds.

- All top models exhibited strong discrimination:
    - Gradient Boosting: 0.8549
    - XGBoost: 0.8533
    - Random Forest: 0.8528
    - Logistic Regression: 0.8527

The narrow spread indicates that while most advanced models discriminate well, AUC-PR and Recall become the decisive differentiators for business usefulness.

Operational Considerations — Training Time

Beyond predictive metrics, operational efficiency is critical for deployment.

- XGBoost trained in 0.387 sec, very fast due to optimized implementation.
- Gradient Boosting required 3.396 sec, still well within operational limits.
- Random Forest (3.828 sec) and AdaBoost (1.687 sec) were moderate.
- Decision Tree and Logistic Regression were fastest but underperformed in predictive quality.

Overall Assessment

Table II Advanced model overall assessment

Model	Key Strengths	Key Limitations
Gradient Boosting	Best AUC-PR, strong recall & ROC-AUC, stable performance	Slightly slower than XGBoost
XGBoost	Excellent AUC-PR & ROC-AUC, very fast training	Slightly lower recall
Random Forest	Highest recall	Lower AUC-PR, slower
AdaBoost	Moderate performance	Weaker AUC-PR
Decision Tree	Simple, fast	Lowest overall predictive quality

Conclusion:
Gradient Boosting and XGBoost clearly dominate the advanced model stage. However, Gradient Boosting provides the best overall trade-off between churn ranking (AUC-PR), churn capture (Recall), discrimination (ROC-AUC), and operational stability, making it the strongest candidate prior to tuning.

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7. Advanced Model Hyperparameter Tuning

7.1 Model selection for tuning

The following three models were selected for tuning:

- Random Forest
- Gradient Boosting
- XGBoost

These models were chosen because:

- They already outperformed simpler models
- They possess tunable hyperparameters that control complexity, generalization, and precision–recall balance
- They are stable under cross-validation

7.2 Tuning strategy

Hyperparameter tuning was explicitly aligned with the business objective and the metric of choice:

- **Scoring Metric:** AUC-PR
- **Objective:** Improve churner ranking and recall–precision balance
- **Validation:** 5-fold cross-validation
- **Search Methods:** RandomizedSearchCV and GridSearchCV

All tuning experiments employed stratified 5-fold cross-validation and were evaluated using AUC-PR as the optimization metric. This ensured that hyperparameter selection directly supported the business objective of maximizing churner identification rather than overall accuracy.

7.3 Performance of Tuned models (Multi-Metric Evaluation)

The tuned models were evaluated across multiple performance metrics — **AUC-PR (primary metric of choice), Recall, F1-score, and ROC-AUC** — to ensure both strong churn ranking and reliable operational behavior.

Table III Tuned models performance metric

Model	AUC-PR (Untuned)	AUC-PR (Tuned)	Recall (Untuned)	Recall (Tuned)	F1 (Untuned)	F1 (Tuned)
XGBoost	0.684	**0.686**	0.784	0.553	0.662	0.609
Random Forest	0.678	**0.683**	0.816	0.758	0.670	0.669
Gradient Boosting	0.686	**0.680**	0.789	0.790	0.671	0.671

Metric-wise Commentary

AUC-PR (Primary Metric of Choice):
XGBoost and Random Forest both show measurable improvement after tuning, indicating stronger churn-risk ranking. Gradient Boosting remains highly competitive with the highest stable AUC-PR even before tuning, demonstrating superior baseline learning quality.

Recall:
Recall remains highest and most stable for Gradient Boosting, confirming its ability to consistently identify churn-prone customers. Although XGBoost achieves a marginal AUC-PR improvement, its recall declines after tuning, reducing its effectiveness for churn prevention.

F1-Score:
Gradient Boosting maintains the strongest and most consistent F1-score, reflecting a reliable balance between precision and recall. Random Forest also maintains stable F1, while XGBoost shows degradation in F1 after tuning.

Overall Assessment

Across all key metrics, Gradient Boosting (Untuned) exhibits the most balanced and operationally reliable performance:

• Highest stable AUC-PR
• Strongest recall behavior
• Consistent F1-score
• Minimal sensitivity to hyperparameter changes

This multi-metric evaluation confirms that Gradient Boosting (Untuned) is the most suitable model for deployment.

7.4 Model validation consistency

To ensure that performance improvements observed during model development were not artifacts of a single train–test split, model training and tuning were conducted using **stratified cross-validation** with five folds. This ensured that the churn distribution was preserved across validation sets and that model evaluation remained stable under varying data partitions.

All tuned models were optimized using **cross-validated AUC-PR** as the primary scoring function, explicitly aligning the optimization process with the business objective of correctly ranking churn-prone customers. The consistency between cross-validation AUC-PR scores and test-set AUC-PR confirms that the selected models generalize reliably and are not overfitting to a particular subset of the data.

Additionally, the use of **decision-threshold optimization** based on the precision–recall curve further strengthened model reliability. Instead of relying on the default probability threshold of 0.5, each model's classification threshold was selected to maximize the F1-score, yielding a more business-appropriate balance between precision and recall. This step significantly improved the stability and practical usefulness of predictions for retention targeting.

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8. Model performance comparison and final model selection

8.1 Final model comparison

Among all baseline, advanced, and tuned models, Gradient Boosting (Untuned) emerged as the most reliable model when evaluated holistically across the project's primary and secondary evaluation metrics.

Table IV Gradient Boost vs XGBoost

Model	Train Time (sec)	AUC-PR	Recall	ROC-AUC	Precision	Accuracy
Gradient Boosting	3.597	0.686	**0.789**	0.855	0.583	0.784
XGBoost (Tuned)	349.059	0.686	0.743	0.856	**0.616**	**0.799**

While both Gradient Boosting (Untuned) and XGBoost (Tuned) achieved identical AUC-PR scores of 0.686, indicating equal ability to rank churn-risk customers, Gradient Boosting demonstrated superior recall and dramatically lower computational cost, making it the strongest overall candidate for deployment.

8.2 Reasoning behind choosing Gradient Boosting (Untuned)

Gradient Boosting (Untuned) was selected as the final model because:

• It achieved the same highest AUC-PR (0.686) as the best tuned alternative
• It delivered higher recall (0.789 vs 0.743), ensuring churners are less likely to be missed
• It maintained strong ROC-AUC (0.855), confirming robust overall discrimination
• It demonstrated exceptional computational efficiency, training in 3.6 seconds compared to 349 seconds for tuned XGBoost
• It avoided the diminishing returns and volatility introduced by aggressive hyperparameter tuning

In addition to superior predictive reliability, Gradient Boosting (Untuned) exhibited stable generalization under cross-validation and consistent behavior after probability-threshold optimization. These properties make it the most operationally dependable choice for real-world deployment.

8.3 Business interpretation of final models

The final model provides AlphaCom with a robust churn-risk ranking engine capable of:

• Identifying high-risk customers early
• Prioritizing retention budgets more effectively
• Supporting pricing, contract, and service-bundling strategies
• Reducing long-term churn-related revenue loss

This model establishes a scalable foundation for continuous learning and future model upgrades as additional customer data becomes available.

8.4 Deployment readiness considerations

The selected model's stability across cross-validation, resistance to overfitting, and consistent decision-threshold behavior indicate strong readiness for operational deployment.

Its probabilistic outputs enable dynamic churn-risk ranking, allowing AlphaCom to continuously reprioritize customers as new behavioral and transactional data is observed.

Moreover, performance metrics remain stable across data partitions, ensuring predictable results when applied to future customer cohorts.

8.5 Final model selection summary

Final chosen model: Gradient Boosting (Untuned)

Table V Final Model Performance Metrics

Metric	Value
AUC-PR	**0.686**
Recall	**0.789**
ROC-AUC	**0.855**
Precision	**0.583**
Accuracy	**0.784**
Training Time	**3.6 seconds**

Why not XGBoost despite tuning?

Although XGBoost (Tuned) achieved the same AUC-PR (0.686), it delivered lower recall and required nearly 100× more training time, offering no material business advantage for the additional computational cost.

Why stability and generalization mattered more than marginal differences:

Given the high cost of missed churners and the need for consistent performance across customer cohorts, model stability, recall, and generalization were prioritized over marginal improvements in secondary metrics. Gradient Boosting (Untuned) provides the optimal balance between predictive power, business impact, and system efficiency.

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9. Key Takeaways

- **Customer churn at AlphaCom is primarily an early-lifecycle challenge.**
  A significant proportion of customers leave within the first year of service, indicating that churn is largely driven by onboarding experience, expectation gaps, and early value perception rather than long-term dissatisfaction. Customers with shorter tenure reassess providers more frequently and are therefore more susceptible to switching, making the initial customer journey a critical retention window.
- **Contract structure is the most influential determinant of churn behavior.**
  Customers on month-to-month contracts exhibit substantially higher churn than those on one-year or two-year agreements. Contractual commitment exerts a powerful stabilizing effect by increasing switching costs, strengthening customer confidence, and reinforcing the long-term service relationship.
- **Pricing sensitivity plays a central role in churn decisions.**
  Higher monthly charges are strongly associated with increased churn risk, especially when combined with short tenure and flexible contracts. Customers are willing to accept higher prices only when perceived service value is clear and consistent; misalignment between cost and perceived benefit significantly accelerates churn.
- **Service value and customer experience materially reduce churn risk.**
  Adoption of value-added services such as **TechSupport** and **OnlineSecurity** is associated with meaningfully lower churn rates, as these services enhance customer engagement, trust, and perceived reliability. In addition, **payment behavior** offers important churn signals, with customers using electronic check payments demonstrating elevated churn, likely due to billing friction or inconvenience.
- **Gradient Boosting emerged as the most reliable and business-aligned predictive model.**
  Among all evaluated models, Gradient Boosting delivered the strongest and most stable performance across AUC-PR, recall, F1-score, and ROC-AUC, while maintaining consistent generalization under cross-validation. Its ability to capture nonlinear relationships between tenure, pricing, contract type, and service usage makes it especially well suited for real-world churn prediction.
- **Model outputs directly enable high-impact retention decision-making.**
  The final model provides AlphaCom with a scalable churn-risk ranking system that supports early identification of vulnerable customers, efficient prioritization of retention resources, and data-driven optimization of contract, pricing, and service strategies.

10. Recommendations

- **Institutionalize early-tenure retention programs as a core business priority.**
  Given that churn is heavily concentrated within the first year of service, AlphaCom should focus retention investments on the first **3–12 months** of the customer lifecycle. This includes structured onboarding, proactive customer success touchpoints, early satisfaction monitoring, and rapid issue resolution. Improvements in this window are likely to generate the highest marginal impact on churn reduction.
- **Use contract design as a strategic retention lever.**
  Customers on month-to-month contracts represent the highest churn risk. AlphaCom should implement targeted upgrade pathways to encourage migration toward longer-term contracts through loyalty incentives, modest pricing advantages, and service bundles. Even partial conversion of this segment can significantly stabilize revenue and reduce churn volatility.
- **Deploy pricing interventions based on churn risk segmentation.**
  Instead of blanket discounts, pricing-sensitive customers should be identified using the model's churn risk scores in combination with monthly charge levels. This enables **personalized retention offers**, protecting margins while maximizing retention effectiveness.
- **Expand early adoption of value-added services.**
  TechSupport and OnlineSecurity should be positioned as core components of the customer value proposition, especially during onboarding and renewal moments. Bundled offerings and trial periods can accelerate adoption, strengthen perceived value, and materially reduce churn risk.
- **Reduce churn driven by billing friction and payment experience.**
  Customers using electronic check payments exhibit higher churn, indicating operational friction rather than service dissatisfaction. AlphaCom should actively promote automated payment methods through convenience incentives and simplified enrollment, thereby removing non-core churn triggers.
- **Operationalize the churn prediction model across business functions.**
  The final Gradient Boosting model should be integrated into marketing, customer success, and pricing workflows as a continuous churn-risk ranking engine. Its outputs can guide retention prioritization, campaign targeting, contract negotiations, and customer experience investments, creating a scalable, data-driven retention framework for long-term competitive advantage.

11. Appendix